Paul Leger: Report from 2025

Associate Professor & Software Engineer
Civil Engineering School
Catholic University of the North
Larrondo 1281, Coquimbo, Chile

Email: pleger@ucn.cl
Homepage: http://pleger.cl
Pragmatics lab: http://pragmaticslab.com (Director)
LinkedIn: http://cl.linkedin.com/in/plegerm

Personal

Name: Paul Leger
Hometown: Coquimbo, Chile
Nationality: Chilean

Publications

My research profile IDs are *ORCID* (0000-0003-0969-5139), *Web of Science* (Q-6174-2017), and *Scopus* (36087550600).

NOTE: For conferences/workshops, I use the CORE ranking, which belongs **CO**mputing **RE**search and Education Association of Austral-asia (near to the publication year). CORE is available on http://portal.core.edu.au/conf-ranks.

Journals (Indexed)

Canquil, Diego, Daniel SanMartín, Irene Inoquio-Renteria, and Paul Leger (Jan. 2026). "A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption". In: *Water Resources Management* 40.2, p. 83. DOI: 10.1007/s11269-026-04496-4. **JCR-WOS**.

Leger, Paul, David Moreno-Lumbreras, Felipe Ruiz, Nicolás Sepúlveda, Ismael Figueroa, and Nicolás Cardozo (Feb. 2026). "The Stackage Repository: An Exploratory Study of its Evolution". In: *Journal of Computer Languages* 86, p. 101386. DOI: 10.1016/j.cola.2026.101386. **JCR-WOS**.

Leger, Paul, Daniel San-Martín, Diego Nuñez, and Tomás Vélez (Feb. 2026). "An Agent-Based Model Reflective Architecture for Social Network Sites". In: *Science of Computer Programming* 252, p. 103452. DOI: 10.1016/j.scico.2026.103452. **JCR-WOS**.

Abdi, Farshid, Shaghayegh Abolmakarem, Amir Karbassi-Yazdi, Paul Leger, Yong Tan, and Giuliani Coluccio (Jan. 2025). "Predicting Patients' Revisit Intention Based on Satisfaction Scores: Combination of Penalized Regression and Neural Networks". In: *IEEE ACCESS* 13, pp. 2783–2800. DOI: 10.1109/ACCESS.2024.3522767. **JCR-WOS**.

Karbassi, Amir, Yang Tan, and Paul Leger (Jan. 2025). "Clustering Financial Institutions in Countries Based on a Hybrid Random Forest and Induced Ordered Weighted Averaging". In: *Journal of Management Analytics*, pp. 1–30. DOI: 10.1080/23270012.2025.2454674. **JCR-WOS**.

Terán, Oswaldo, Paul Leger, and Manuela López (Apr. 2025). "Factors that Drive Market Share and the Oligopolistic Character of Cross-border B2C Ecommerce: An agent-based scenario analysis approach". In: *Simulation: Transactions of the Society for Modeling and Simulation International* 0.0, pp. 1–24. DOI: 10.1177/00375497241296542. **JCR-WOS**.

Conferences (SCOPUS)

Liu, Shijin, Hiroaki Fukuda, and Paul Leger (Mar. 2026). "SOM-based Continuous XGBoost Approach for DDoS Attack Detection and Mitigation in Edge SDN". In: *Workshop on Programmable and Learning Architectures for Intelligent Networks*. Athens, Greece. **SCOPUS. Rank National-France (CORE ranking). To appear**.

Ogino, Toshihiro, Hiroaki Fukuda, and Paul Leger (July 2025). "Stepwise Distributed Execution for Load Reduction in Machine Learning Systems". In: *International Symposium on Parallel and Distributed Computing (ISPDC)*. Rennes, France. **To appear. SCOPUS. Rank C (CORE ranking)**.

You, Koshi, Hiroaki Fukuda, and Paul Leger (Apr. 2025). "The Impact of Context-Oriented Programming on Declarative UI Design in React". In: *Evaluation of Novel Approaches to Software Engineering (ENASE)*. Porto, Portugal, pp. 700–707. DOI: 10.5220/0013432300003928. **SCOPUS. Rank B (CORE ranking)**.

Workshops (SCOPUS)

Congresses (Conferences/Workshops not indexed)

Note: These papers are just presentations of technical reports or working papers.

Moreno-Lumbreras, David, Paul Leger, Sergio Montes-León, Jesus Gonzalez-Barahona, and Gregorio Robles (Sept. 2025). "Evaluating the Comprehension of the Stackage Ecosystem: A Comparison between VR and 2D Visualizations". In: *Registered Report: 2025 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. Auckland, New Zealand. DOI: 10.2139/ssrn.5362665. **Registered Report**.

Posters

Scientific Services

Manager (*x 2*)

- Director on the Engineering School at UCN, Coquimbo (Chile), 2023 - present, https://www.ucn.cl.

- Pragmatics Lab. Founder and director of the research laboratory in software development and simulations of agent-based models, Virtual (Chile), 2018 - present, https://pragmaticslab.com.

Journal Editor (*x 3*)

- Associate Editor on Frontiers in Computer Science, ISSN: 2624-9898, Frontiers, 2025 - present, **JCR-WOS**.

- Editorial Member on Journal of Innovations in Digital Marketing, ISSN: 2765-8341, Luminous Insights, 2021 - present.

- Guest Editor on Frontiers in Computer Science - Special Issue on Artificial Intelligence for Software Engineering: Advances, Applications, and Implications, ISSN: 2624-9898 - Frontiers, **2025, NOTE:** 2025.

Journal Reviewer (*x 1*)

- Journal of Systems and Software, ISSN: 0164-1212, ELSEVIER, 2025, **JCR-WOS**.

Program Committee Member (*x 5*)

- International Conference on Software Technologies (ICSOFT), Porto (Portugal), 2026, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

- (SPLASH) 2025 Workshop on Virtual Machines and Language Implementations, Singapore, 2025, **SCOPUS**, https://conf.researchr.org/home/icfp-splash-2025/vmil-2025.

- International Conference on Software Maintenance and Evolution (ICSME) - Industry Track, Auckland (New Zealand), 2025, **SCOPUS**, https://conf.researchr.org/track/icsme-2025/icsme-2025-industry-track.

- XXXVI International Marketing Congress (AEMARK), Murcia (Spain), 2025, https://aemarkcongresos.com/areas-tematicas-y-comite-cientifico.

- International Conference on Software Technologies (ICSOFT), Bilbao (Spain), 2025, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

Evaluator (*x* 5)

- President of the Ph.D thesis commission: "Diversidad de Licencias en Software Libre", Universidad Rey Juan Carlos (Spain), 2025, https://en.urjc.es.

- President of the Ph.D thesis commission: "Integrated Mathematical Models for Agricultural Decision-Making Under Uncertainty: Efficiency; Water Optimization and Resilience in Olive Production.", Universidad Católica del Norte (Chile), 2025, https://www.ucn.cl.

- Evaluator of Fondef-IDeA Projects (**x 3**), ANID (Chile), 2023 - 2025, https://anid.cl/concursos/concurso-idea-id-tecnologias-avanzadas-2025/2025.

Software Development for Research

My software projects are more related to creating complex software artifacts than developing traditional industrial software (*e.g.,* Inventory Management). Source: http://pleger.cl/software.

PL/SE research with AI: Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), and EMA-Kotlin (adaptation from EMA-js). https://pleger.github.io/PL-SE-research-with-AI, 2026. [**ARTIFICIAL INTELLIGENCE**].

Academic Experience

Undergraduate (*x* 12)

Complex Engineering Problems (*x* 4), Information and Information Technology, Catholic University of the North, 2024 - 2026.

Programming Languages (*x* 8), Computer Civil Engineering, Catholic University of the North, 2021 - 2025.

Thesis Advising (*x* 27)

Postgraduate (*x* 3), Ph. D and Master, Catholic University of the North, 2017 - 2025.

Undergrad (*x* 24), Engineering, Catholic University of the North, 2013 - 2025.

Last updated: April 4, 2026
This resume is available on `http://pleger.cl/resume`